SKU-Level Demand Forecasting Framework for A Large European Retailer

The Trend Setters

IEOR E4578: Forecasting: A Real-World Application

March 2026

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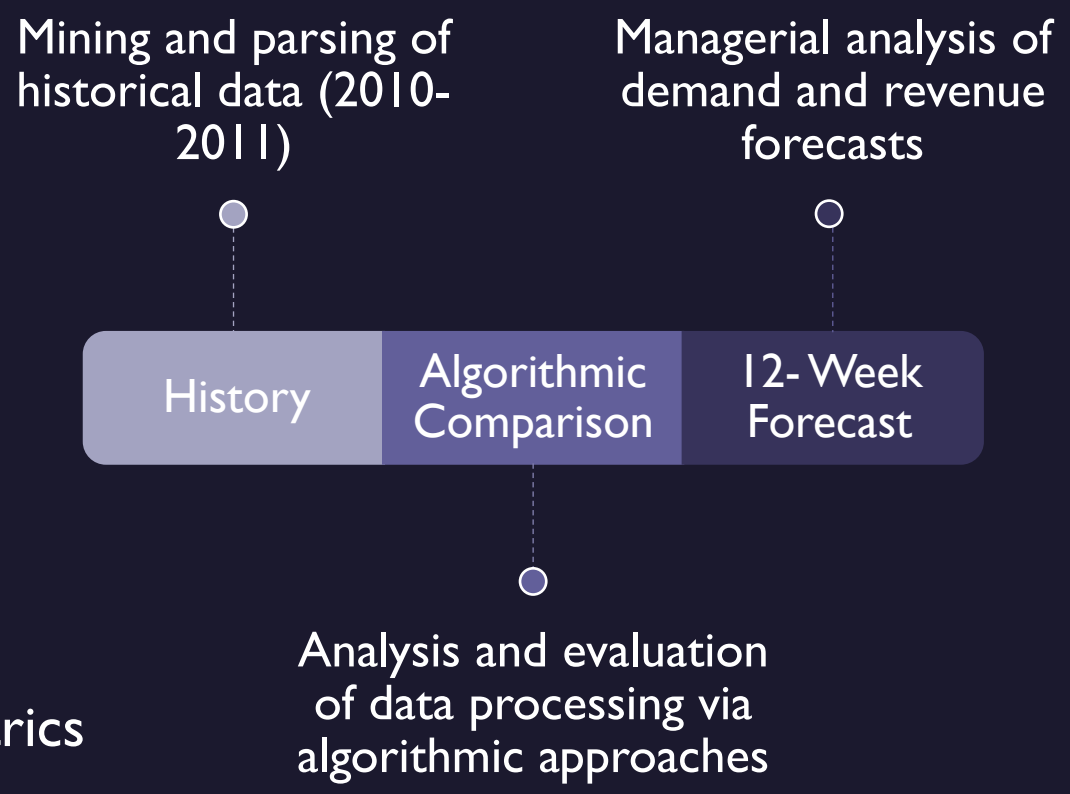
Problem & Objective

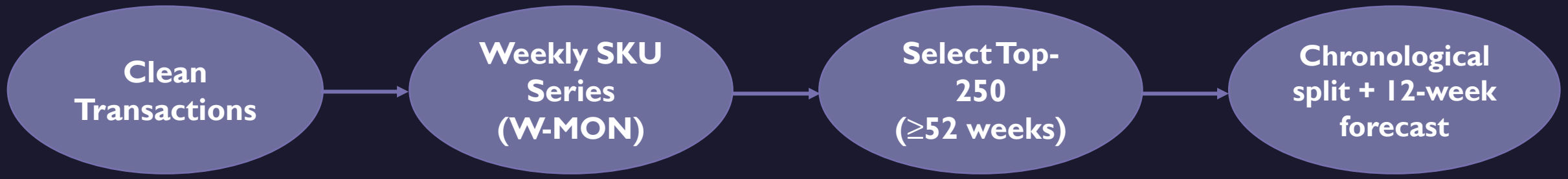
Our analysis aims to assist in **evaluating** the retailer's **operational outlook over the next quarter**

- Forecast 12-week ahead SKU-level demand
- Use **train / validation / test**
- Select model per SKU using validation block-MAPE
- Report held-out test MAPE
- Divide test into **4 equal 4-week periods**
- Translate forecasts into revenue + concentration metrics

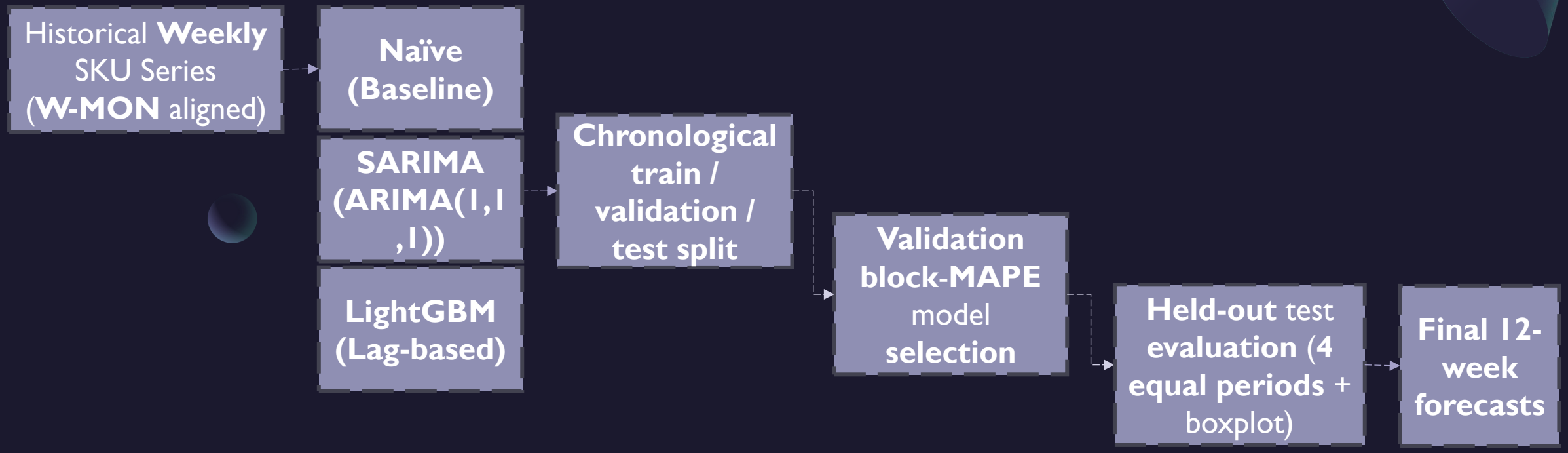


Data & Evaluation Universe

- **Data cleaning:** Exclude cancellations and non-positive transactions (Quantity ≤ 0, Price ≤ 0)
- **Weekly series**: Align to **Monday weeks (W-MON)** to avoid calendar misalignment
- **Screen**: **Top 250 SKUs by revenue with enough history for train + validation + test**
- **Minimum required history = 52 weeks**
- Final **28 weeks** reserved for **out-of-sample** evaluation: **12 validation** weeks + **16 test** weeks
- **Final operational deliverable: 12-week forecast**s



Modelling Framework

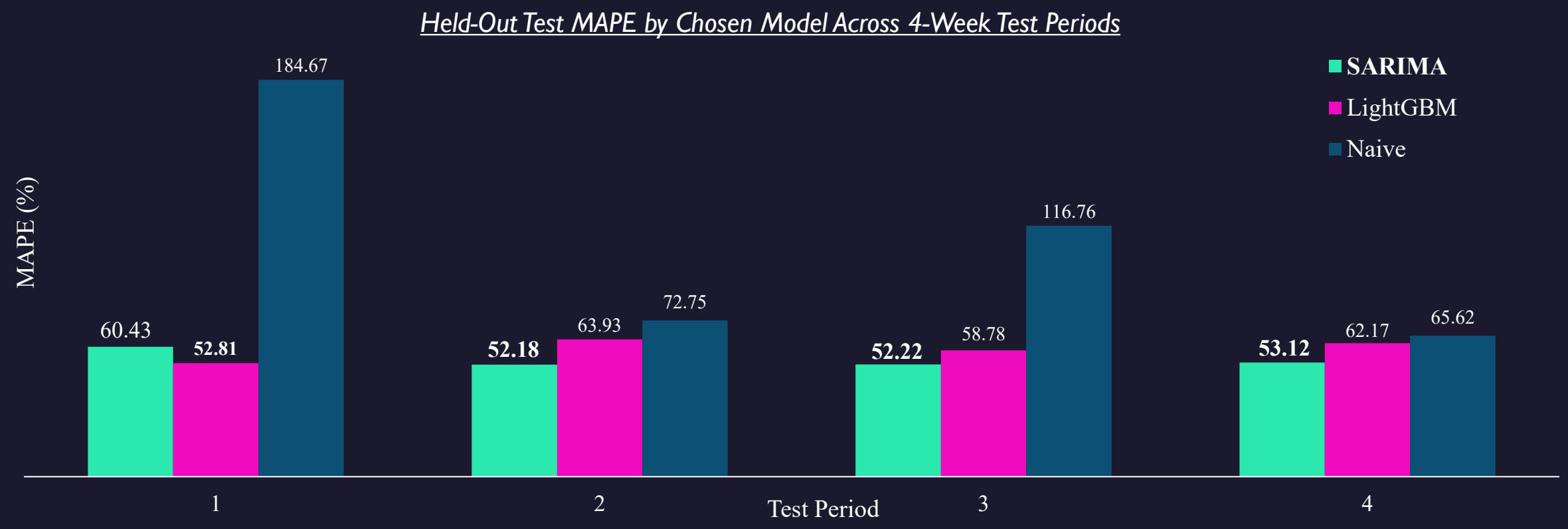


Conceptual diversity improves **robustness** across SKU volatility regimes

Naïve	SARIMA (ARIMA(1,1,1))	LightGBM
✓ Stability benchmark	✓ Models autocorrelation	✓ Nonlinear lag effects
✓ Low variance	✓ Structured & interpretable	✓ Flexible across SKUs
✕ Ignores structure	✕ Assumes linear dynamics	✕ Higher variance

Held-Out Test Performance Across Periods

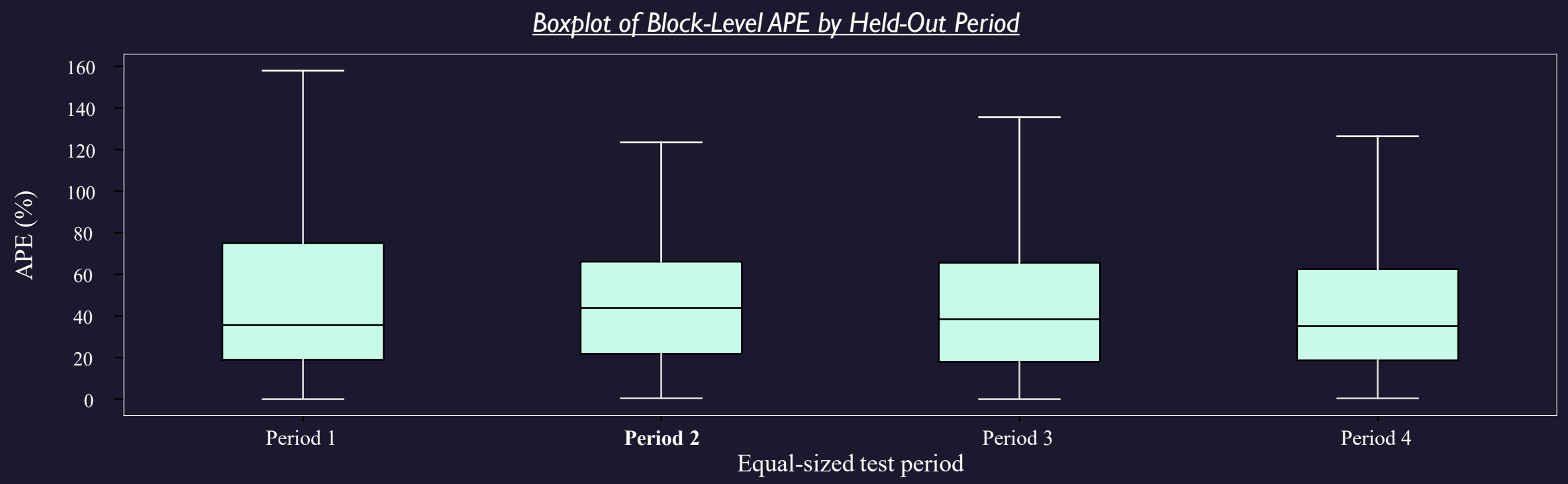
Block-level test MAPE across four equal-sized 4-week held-out periods



- **SARIMA** is the **most stable** chosen-model group **(52–60 MAPE across all four periods).**
- **Naïve** is the **least robust, spiking** to **184.67** in **Period 1** and **116.76** in **Period 3; LightGBM** sits **in between (53–64)**.

Error Spread Across Periods

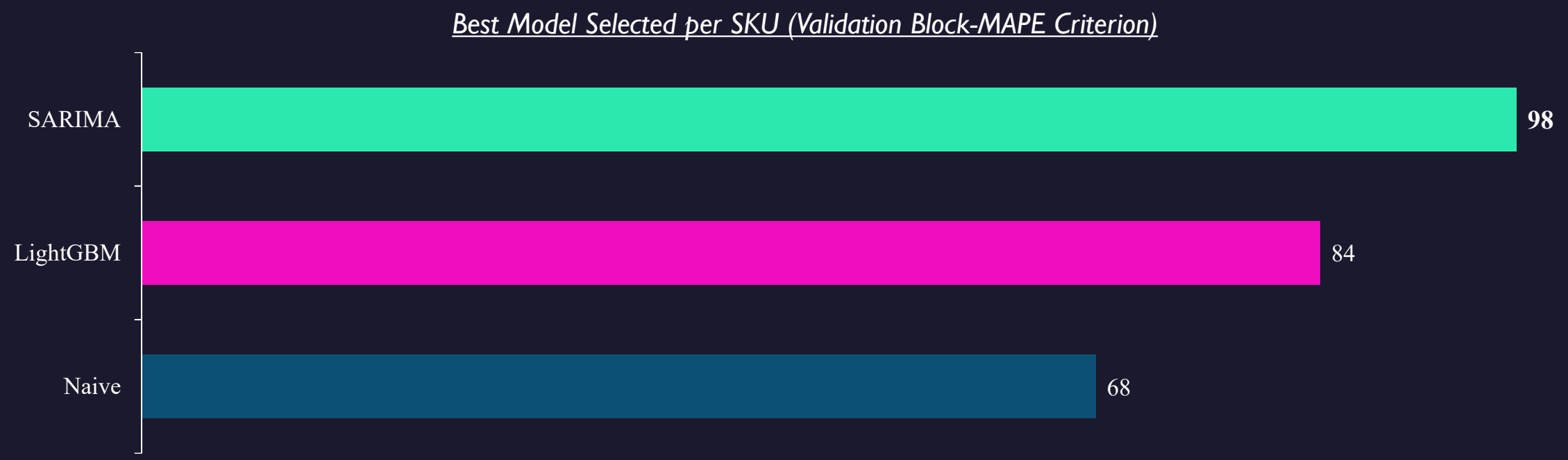
Distribution of block-level APE across four equal-sized 4-week held-out periods



- **Median error** stays **stable across periods (35–44 MAPE); variation** comes **mainly from upper-tail risk.**
- **Period 1** is the **hardest test window; Period 4** is the **most favorable.**

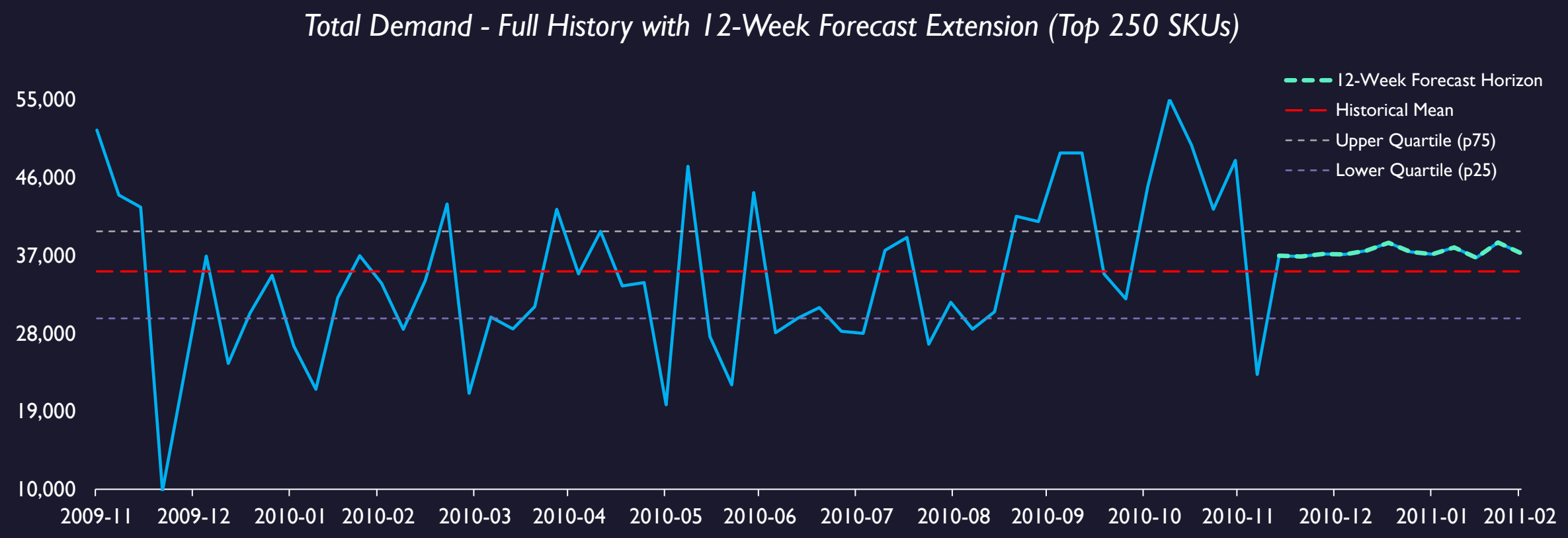
Model Selection Outcomes

Best model frequency under the validation block-MAPE criterion



- **SARIMA** is selected **most often (98/250 SKUs; 39.2%),** confirming it as the **strongest overall specification** in the **panel.**
- **LightGBM** remains competitive for a substantial **minority (84 SKUs; 33.6%)**, while **Naive** is selected **least often** (**68 SKUs; 27.2%**).

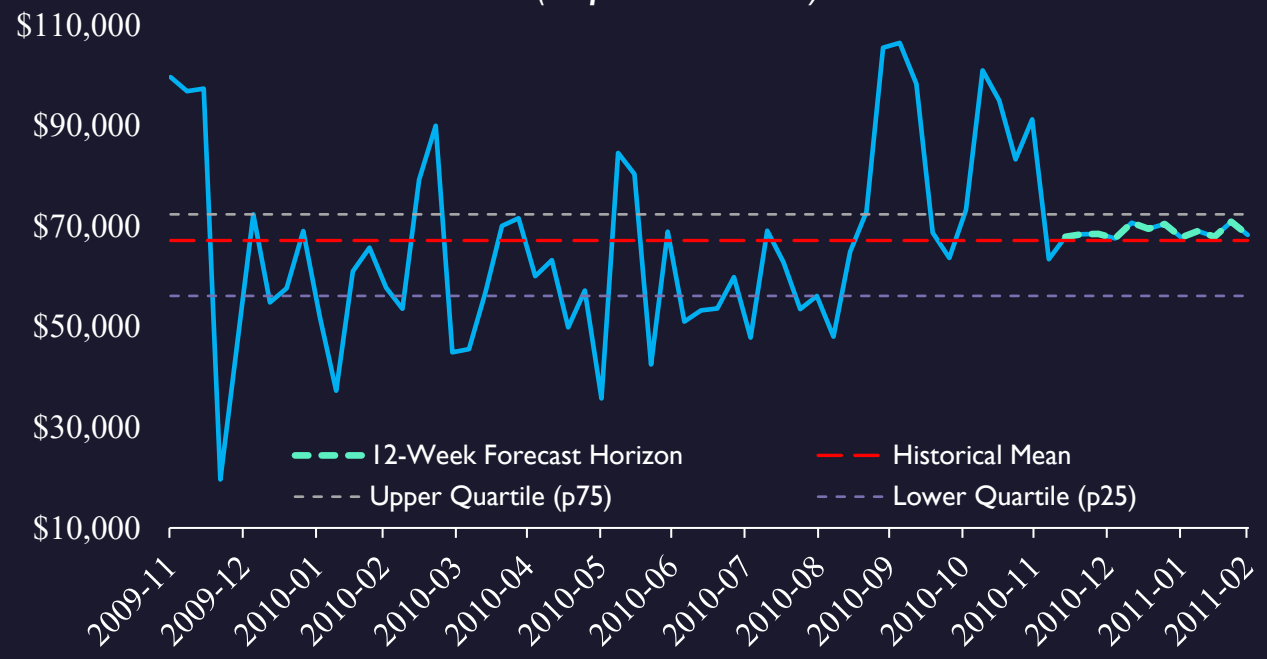
Aggregate Demand Forecast (12-Week Outlook)



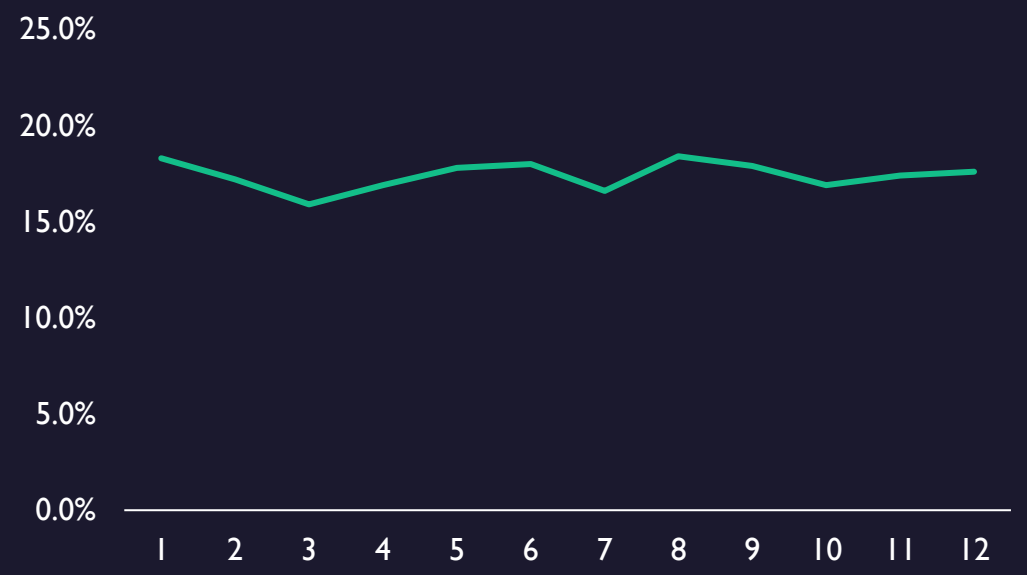
- **Aggregate demand** remains **within historical interquartile range** and modestly **above long-run mean** into **12-week horizon (c. 36k–39k).**
- No evidence of structural break or upward volatility shift in projected total.

Revenue Outlook & Concentration (12-Week Horizon)

Total Revenue - Full History with 12-Week Forecast Extension (Top 250 SKUs)



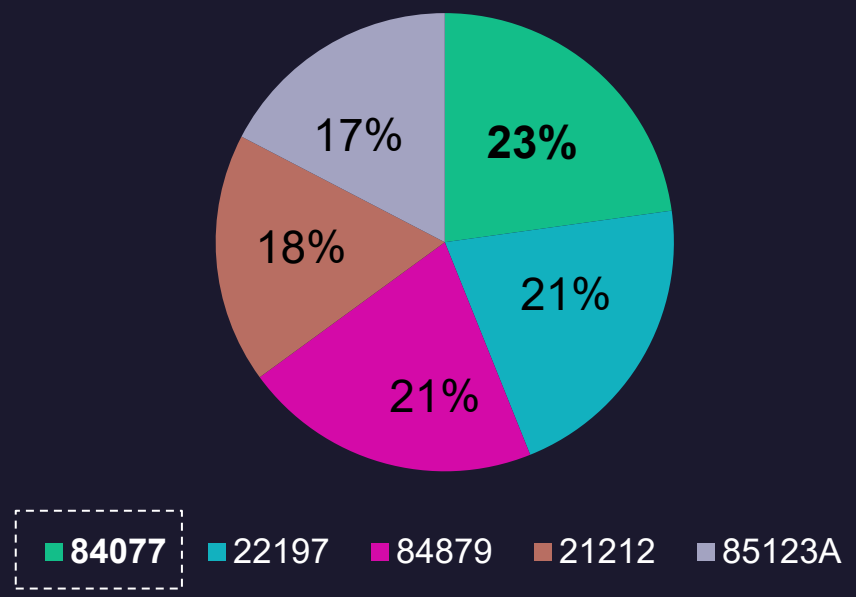
Top-5 Share of Total Revenue Per Horizon



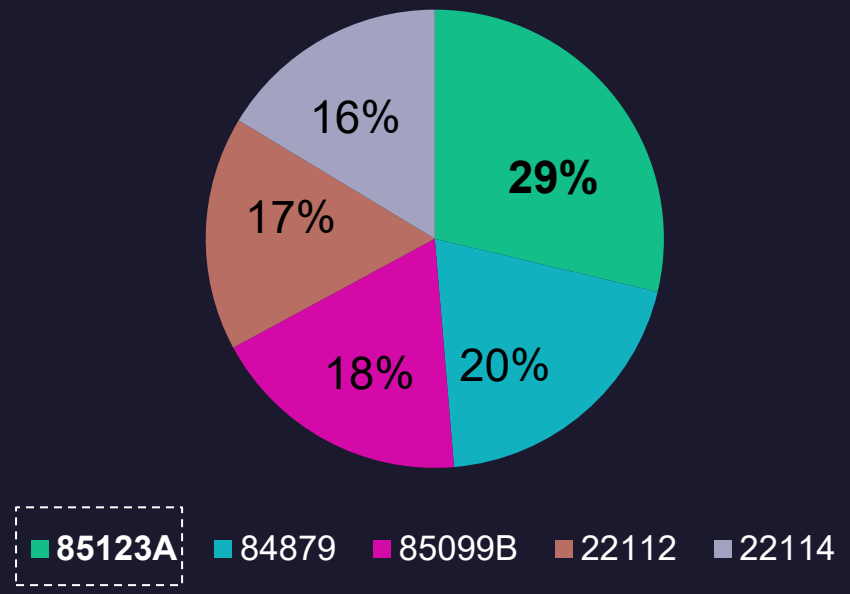
- **Projected revenue** remains modestly **above long-run historical mean,** reinforcing a **stable short-term outlook ($67k–$72k).**
- **Top-5 SKUs** account for **~16–19% of projected revenue,** with **no upward concentration trend across horizons.**

Top-Tier SKUs: Revenue Composition

Top 5 SKUs By Projected Demand (H=12)



Top 5 SKUs By Projected Revenue (H=12)

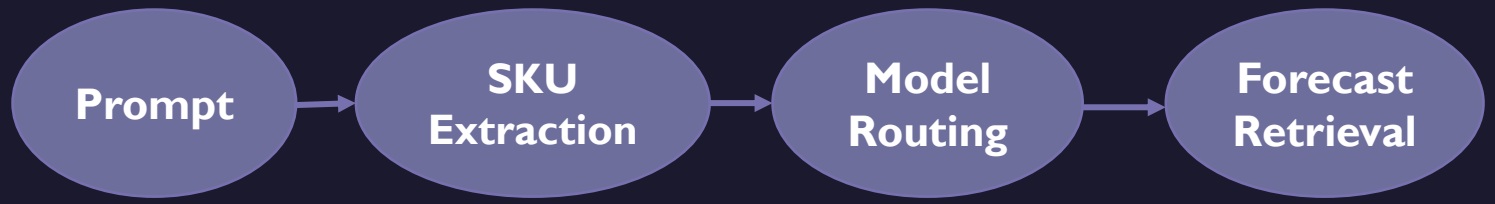


- **Top contributor rotates** between demand and revenue, implying different operational priorities: **volume planning vs. profit protection**.
- **Revenue shares** are **more uneven** than **demand shares,** suggesting **pricing / margin heterogeneity** across **top SKUs.**

Agentic Forecast Retrieval Layer

Purpose: Enable business users to retrieve the latest **SKU-specific 12-week forecast** through a **natural-language prompt**.

- **User enters** a **natural-language prompt** containing a valid SKU
- **Interface extracts** the **SKU** and **routes** to the **selected model**
- Latest **12-week demand** and **revenue forecast** is **returned**.



Example output from the prompt-based retrieval interface for two valid SKU requests

```
40]: example_prompt_1 = "What is the latest 12-week forecast for product 22197?"
     example_prompt_2 = "What is the 12-week demand and revenue outlook for product 84077?"

     print(get_latest_forecast_for_sku(example_prompt_1, agent_summary, agent_horizon))
     print()
     print(get_latest_forecast_for_sku(example_prompt_2, agent_summary, agent_horizon))

Forecast retrieval result for SKU 22197
--------------------------------------------------
Selected model: SARIMA
Validation block-MAPE: 27.63
12-week projected demand: 16977.8
12-week projected revenue: $14,431.16
Median historical price proxy: $0.85

Weekly forecast breakdown:
Week  1: Demand = 1499.6, Revenue = $1,274.63
Week  2: Demand = 1392.9, Revenue = $1,183.97
Week  3: Demand = 1410.9, Revenue = $1,199.28
Week  4: Demand = 1407.9, Revenue = $1,196.69
Week  5: Demand = 1408.4, Revenue = $1,197.13
Week  6: Demand = 1408.3, Revenue = $1,197.06
Week  7: Demand = 1408.3, Revenue = $1,197.07
Week  8: Demand = 1408.3, Revenue = $1,197.07
Week  9: Demand = 1408.3, Revenue = $1,197.07
Week 10: Demand = 1408.3, Revenue = $1,197.07
Week 11: Demand = 1408.3, Revenue = $1,197.07
Week 12: Demand = 1408.3, Revenue = $1,197.07

Forecast retrieval result for SKU 84077
--------------------------------------------------
Selected model: SARIMA
Validation block-MAPE: 20.55
12-week projected demand: 18325.6
12-week projected revenue: $3,848.38
Median historical price proxy: $0.21

Weekly forecast breakdown:
Week  1: Demand = 1653.3, Revenue = $347.19
Week  2: Demand = 1501.0, Revenue = $315.22
Week  3: Demand = 1518.8, Revenue = $318.95
Week  4: Demand = 1516.7, Revenue = $318.52
Week  5: Demand = 1517.0, Revenue = $318.57
Week  6: Demand = 1517.0, Revenue = $318.56
Week  7: Demand = 1517.0, Revenue = $318.56
Week  8: Demand = 1517.0, Revenue = $318.56
Week  9: Demand = 1517.0, Revenue = $318.56
Week 10: Demand = 1517.0, Revenue = $318.56
Week 11: Demand = 1517.0, Revenue = $318.56
Week 12: Demand = 1517.0, Revenue = $318.56
```

Managerial Takeaways

- Near-term demand and revenue remain stable and modestly above historical averages, supporting a steady 12-week planning horizon.
- Revenue leadership differs from volume leadership, supporting targeted pricing and margin optimisation across top-tier SKUs (e.g., 85123A).
- SARIMA demonstrates the most robust cross-SKU performance, offering a scalable default model when operational simplicity is prioritised.
- Next value lever: incorporate exogenous drivers (pricing, promotions, holidays) to improve predictive power beyond historical demand patterns.

Modeling Limitatons and Next Steps

Limitations	Next Steps
Limited historical depth, constraining full seasonal decomposition.	**Extend historical coverage** to strengthen seasonal structure estimation.
No exogenous drivers, so forecasts rely solely on internal demand dynamics.	Integrate **external drivers** such as promotions, pricing, holidays, and macro factors.
Recursive multi-step forecasting, introducing potential error propagation at longer horizons.	Explore direct **multi-horizon** or probabilistic forecasting frameworks.
Static price assumption in revenue translation.	**Stress-test revenue** under **alternative price scenarios**.

Summary

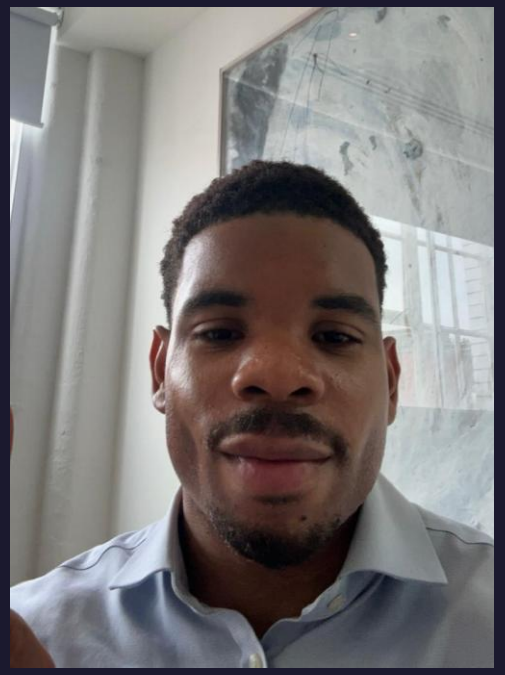
Forecasts exhibit **stable, mean-reverting demand dynamics** with **no structural break** over the 12-week horizon.

Revenue projections remain modestly **above long-run averages without** introducing concentration **risk**.

The 12-week planning window provides a **statistically defensible** and **operationally practical** framework for short- to medium-term inventory decisions.

Team

Ishmeen Garewal



Aka Oguine

Khushi Advani